

Icelandic with Compose Keys on Linux

Rohit Goswami · 2024-03-08 · workflow · tools · keyboard · linux

Minimally expanding my existing layouts for supporting Icelandic orthography.^[^fn:1]

Background

Life outside Emacs can be daunting for a lot of reasons, the lack of `auctex` and `org-latex` being a major one. Additionally, for the purposes of my (somewhat current) country of residence^[^fn:2], it is often useful to be able to type Icelandic characters without resorting to something like `typeit`.

Compositional Choices

I already use `klfc` to generate a very portable (in a Unix sense) customized [VIM based Colemak binding layout](#). Some reasons why I first decided to modify the compose keybindings instead of extending that are^[^fn:3]:

- The keys and bindings defined here aren't in use enough for me to justify actually adding them to one of my levels.
- It is slightly easier to share `.XCompose` files compared to the `klfc` compatible `json` files.
- It also integrates nicely with [my dotfiles](#) and templates
 - There are a few tools for concatenating fragments^[^fn:4] but it seemed more straightforward to use my existing `chezmoi` machinery
 - The tools, especially [ext-compose](#), can be used as a pretty definitive source of viable mappings

However, though the compose key and chords works great for TeX it isn't exactly ideal for language bindings, so I ended up with a hybrid, where some keys are placed on the dead-greek (`AltGr`) level and the rest are via multi-key (a.k.a `compose`) chords.

Setting the Compose Key

Modern keyboards don't often come with a dedicated `compose` key. For simplicity I opted to choose one of the default supported rules for the this:

```

grep "compose:" /usr/share/X11/xkb/rules/base.lst
compose:ralt      Right Alt
compose:lwin      Left Win
compose:lwin-altgr 3rd level of Left Win
compose:rwin      Right Win
compose:rwin-altgr 3rd level of Right Win
compose:menu      Menu
compose:menu-altgr 3rd level of Menu
compose:lctrl     Left Ctrl
compose:lctrl-altgr 3rd level of Left Ctrl
compose:rctrl     Right Ctrl
compose:rctrl-altgr 3rd level of Right Ctrl
compose:caps      Caps Lock
compose:caps-altgr 3rd level of Caps Lock
compose:l02       The "&lt; &gt;" key
compose:l02-altgr 3rd level of the "&lt; &gt;" key
compose:paus      Pause
compose:ins       Insert
compose:prsc      PrtSc
compose:sclk      Scroll Lock

```

I went with `compose:ralt` since `compose:lalt` isn't thing. Setting this in `sway` is simple and stacks well with the other settings:

```

# To test
swaymsg 'input * xkb_options compose:ralt'

```

To set this permanently:

```

input * {
    xkb_options "compose:ralt"
}

```

Mapping Icelandic

I used [TypeIt](#) as a reference, since I'm rather used to it. The mapping therefore proceeds in two stages, an AltGr layer, and some compose keychords.

AltGr Layer

For the base keys, I did end up first extending the AltGr layer, with the following, as depicted in Fig.

[fig:altgrlayeris](#) [^fn:5]:

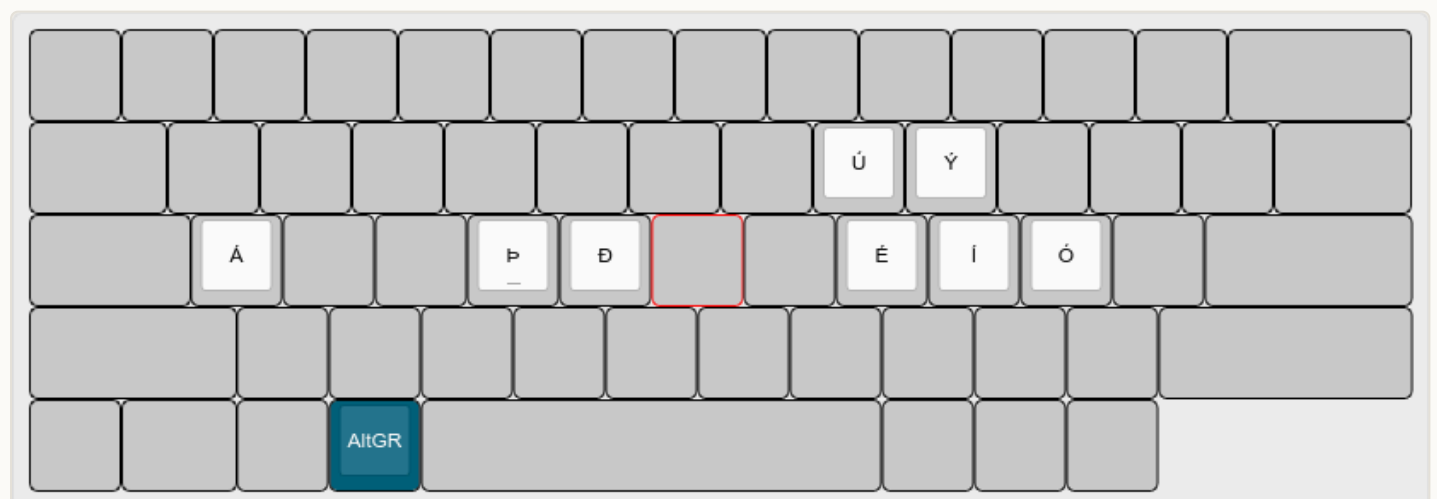


Figure 1: AltGr layer mapping

Which is defined in another `json` file:

```
{
  "filter": "no tmk",
  "singletonKeys": [
    [ "Alt_L", "AltGr" ]
  ],
  "shiftlevels": [ "AltGr", "Shift+AltGr" ],
  "keys": [
    { "pos": "A", "letters": [ "á", "Á" ] },
    { "pos": "F", "letters": [ "þ", "Þ" ] },
    { "pos": "G", "letters": [ "ð", "Ð" ] },
    { "pos": "K", "letters": [ "é", "É" ] },
    { "pos": "L", "letters": [ "í", "Í" ] },
    { "pos": ";", "letters": [ "ó", "Ó" ] },
    { "pos": "I", "letters": [ "ú", "Ú" ] },
    { "pos": "O", "letters": [ "ý", "Ý" ] }
  ]
}
```

Installation is similar to the previous post:

```
git clone https://github.com/39aldo39/klfc
cd klfc
# Kanged from the AUR https://aur.archlinux.org/packages/klfc/
cabal v1-update
cabal v1-install --only-dependencies --ghc-options=-dynamic --force-reinstalls
cabal v1-configure --prefix=/usr --ghc-options=-dynamic
cabal v1-build
# Build layout
./dist/build/klfc/klfc ~/.config/klfc/colemak.json ~/.config/klfc/extendVIM.json
~/.config/klfc/is_altgr_Colevim.json -o colevimIS
# Install
cd colevimIS/xkb/
./install-system.sh && ./scripts/install-xcompose.sh
```

Compose Keychords

It turns out there are really only two left:

```
# ~/.XCompose
<Multi_key> <a> <e> : "æ"
<Multi_key> <o> <i> : "ö"
```

Conclusions

This is going to take a while to get used to for sure, but eventually it should help me reduce the time it takes me to get thoughts on (digital) paper in Icelandic. I will probably need to touch type on something like [fastfingers](#) [^fn:6].

I had hoped to include a bunch of unicode math and other compose keychords, but it is annoying to have to keep track of the translations like `*` to `asterisk` and similar[^fn:7].